Лабораторная работа №14

Статическая маршрутизация в Интернете. Настройка

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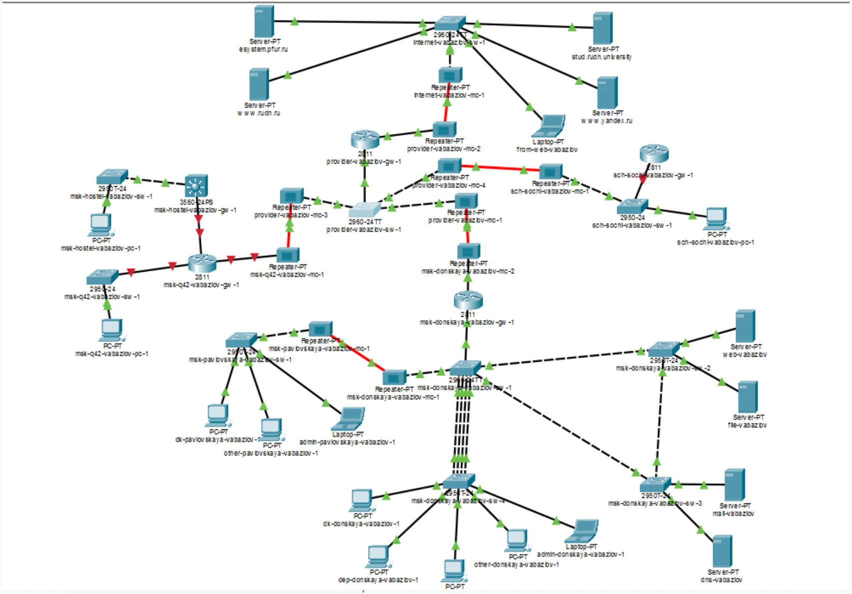
Цель работы

Общая топология сети

Схема сети



Server-PT
esystem.pfur.ru
2960-24TT
internet-vabazlov-sw-1
Server-PT
stud.rudn.university
Server-PT
www.rudn.ru
Repeater-PT
internet-vabazlov-mc-1
Server-PT
www.yandex.ru
2811
provider-vabazlov-gw-1
Repeater-PT
provider-vabazlov-mc-2
Laptop-PT
from-web-vabazlov
2811
sch-sochi-vabazlov-gw-1
2950T-24
msk-hostel-vabazlov-sw-1
3560-24PS
msk-hostel-vabazlov-gw-1
Repeater-PT
provider-vabazlov-mc-4
Repeater-PT
sch-sochi-vabazlov-mc-1
Repeater-PT
provider-vabazlov-mc-3
2960-24TT
provider-vabazlov-sw-1
Repeater-PT
provider-vabazlov-mc-1
2950-24
sch-sochi-vabazlov-sw-1
PC-PT
sch-sochi-vabazlov-pc-1
PC-PT
msk-hostel-vabazlov-pc-1
2811
msk-q42-vabazlov-gw-1
Repeater-PT
msk-q42-vabazlov-mc-1
Repeater-PT
msk-donskaya-vabazlov-mc-2
2950-24
msk-q42-vabazlov-sw-1
2811
msk-donskaya-vabazlov-gw-1
PC-PT
msk-q42-vabazlov-pc-1
Repeater-PT
msk-pavlovskaya-vabazlov-mc-1
2950T-24
msk-pavlovskaya-vabazlov-sw-1
Server-PT
web-vabazlov
2950T-24
msk-donskaya-vabazlov-sw-2
Repeater-PT
msk-donskaya-vabazlov-mc-1
msk-donskaya-vabazlov-sw-1
Server-PT
file-vabazlov
PC-PT
dk-pavlovskaya-vabazlov-1
PC-PT
other-pavlovskaya-vabazlov-1
Laptop-PT
admin-pavlovskaya-vabazlov-1
2950T-24
msk-donskaya-vabazlov-sw-4
2950T-24
msk-donskaya-vabazlov-sw-3
Server-PT
mail-vabazlov
PC-PT
dk-donskaya-vabazlov-1
PC-PT
dep-donskaya-vabazlov-1
PC-PT
PC-PT
other-donskaya-vabazlov-1
Laptop-PT
admin-donskaya-vabazlov-1
Server-PT
dns-vabazlov

Основные элементы

Настройка сети провайдера

VLAN и trunk

```
provider-vabazlov-sw-1>en
Password:
provider-vabazlov-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-vabazlov-sw-1(config)#int f0/3
provider-vabazlov-sw-1(config-if)#sw mode trunk

provider-vabazlov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

provider-vabazlov-sw-1(config-if)#int f0/4
provider-vabazlov-sw-1(config-if)#sw mode trunk
provider-vabazlov-sw-1(config-if)#vlan 5
provider-vabazlov-sw-1(config-vlan)#name q42
provider-vabazlov-sw-1(config-vlan)#int vlan 5
provider-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up

provider-vabazlov-sw-1(config-if)#no sh
provider-vabazlov-sw-1(config-if)#vlan 6
provider-vabazlov-sw-1(config-vlan)#name sochi
provider-vabazlov-sw-1(config-vlan)#int vlan 6
provider-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-vabazlov-sw-1(config-if)#no sh
provider-vabazlov-sw-1(config-if)#exit
provider-vabazlov-sw-1(config)#exit
provider-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-vabazlov-sw-1#wr mem
Building configuration...
[OK]
provider-vabazlov-sw-1#
```

Настройка центрального шлюза

Подинтерфейсы и маршруты

```
nsk-donskaya-vabazlov-gw-1(config)#
nsk-donskaya-vabazlov-gw-1(config)#int f0/1.5
nsk-donskaya-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

nsk-donskaya-vabazlov-gw-1(config-subif)#enc dot1q 5
nsk-donskaya-vabazlov-gw-1(config-subif)#ip addr 10.128.255.1 255.255.255.252
nsk-donskaya-vabazlov-gw-1(config-subif)#desc q42
nsk-donskaya-vabazlov-gw-1(config-subif)#int f0/1.6
nsk-donskaya-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up

nsk-donskaya-vabazlov-gw-1(config-subif)#enc dot1q 6
nsk-donskaya-vabazlov-gw-1(config-subif)#ip addr 10.128.255.5 255.255.255.252
nsk-donskaya-vabazlov-gw-1(config-subif)#descr sochi
nsk-donskaya-vabazlov-gw-1(config-subif)#exit
nsk-donskaya-vabazlov-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.2
nsk-donskaya-vabazlov-gw-1(config)#ip route 10.130.0.0 255.255.0.0 10.128.255.6
nsk-donskaya-vabazlov-gw-1(config)#int f0/1.5
nsk-donskaya-vabazlov-gw-1(config-subif)#ip nat inside
nsk-donskaya-vabazlov-gw-1(config-subif)#int f0/1.6
nsk-donskaya-vabazlov-gw-1(config-subif)#ip nat inside
nsk-donskaya-vabazlov-gw-1(config-subif)#ip access-list ext nat-inet
nsk-donskaya-vabazlov-gw-1(config-ext-nacl)#remark q42
nsk-donskaya-vabazlov-gw-1(config-ext-nacl)#permit ip host 10.129.0.200 any
nsk-donskaya-vabazlov-gw-1(config-ext-nacl)#permit ip host 10.129.128.200 any
nsk-donskaya-vabazlov-gw-1(config-ext-nacl)#remark sochi
nsk-donskaya-vabazlov-gw-1(config-ext-nacl)#permit ip host 10.130.0.200 any
nsk-donskaya-vabazlov-gw-1(config-ext-nacl)#exit
nsk-donskaya-vabazlov-gw-1(config)#exit
nsk-donskaya-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

nsk-donskaya-vabazlov-gw-1#wr mem
Building configuration...
[OK]
nsk-donskaya-vabazlov-gw-1#
```

Настройка сети 42-го квартала

Маршрутизатор 42-го квартала

```
msk-q42-vabazlov-gw-1(config-if)#int f0/1.5
msk-q42-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-q42-vabazlov-gw-1(config-subif)#enc dot1q 5
msk-q42-vabazlov-gw-1(config-subif)#ip addr 10.128.255.2 255.255.255.252
msk-q42-vabazlov-gw-1(config-subif)#descr donskaya
msk-q42-vabazlov-gw-1(config-subif)#int f0/0
msk-q42-vabazlov-gw-1(config-if)#no sh

msk-q42-vabazlov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-q42-vabazlov-gw-1(config-if)#int f0/0.201
msk-q42-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.201, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up

msk-q42-vabazlov-gw-1(config-subif)#enc dot1q 201
msk-q42-vabazlov-gw-1(config-subif)#ip addr 10.129.0.1 255.255.255.0
msk-q42-vabazlov-gw-1(config-subif)#desc q42-main
msk-q42-vabazlov-gw-1(config-subif)#int f1/0
msk-q42-vabazlov-gw-1(config-if)#no sh

msk-q42-vabazlov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

msk-q42-vabazlov-gw-1(config-if)#int f1/0.202
msk-q42-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet1/0.202, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up

msk-q42-vabazlov-gw-1(config-subif)#enc do1q 202
                                          ^
% Invalid input detected at '^' marker.

msk-q42-vabazlov-gw-1(config-subif)#enc dot1q 202
msk-q42-vabazlov-gw-1(config-subif)#ip addr 10.129.1.1 255.255.255.0
msk-q42-vabazlov-gw-1(config-subif)#desc q42-management
msk-q42-vabazlov-gw-1(config-subif)#exit
msk-q42-vabazlov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.1
msk-q42-vabazlov-gw-1(config)#ip route 10.129.128.0 255.255.128.0 10.129.1.2
msk-q42-vabazlov-gw-1(config)#exit
```

Коммутатор 42-го квартала

```
msk-q42-vabazlov-sw-1(config)#
msk-q42-vabazlov-sw-1(config)#int f0/24
msk-q42-vabazlov-sw-1(config-if)#sw mode trunk

msk-q42-vabazlov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

msk-q42-vabazlov-sw-1(config-if)#int f0/1
msk-q42-vabazlov-sw-1(config-if)#sw mode access
msk-q42-vabazlov-sw-1(config-if)#sw acc vlan 201
% Access VLAN does not exist. Creating vlan 201
msk-q42-vabazlov-sw-1(config-if)#vlan 201
msk-q42-vabazlov-sw-1(config-vlan)#name q42-main
msk-q42-vabazlov-sw-1(config-vlan)#int vlan 201
msk-q42-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up

msk-q42-vabazlov-sw-1(config-if)#no sh
msk-q42-vabazlov-sw-1(config-if)#exit
msk-q42-vabazlov-sw-1(config)#exit
msk-q42-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-vabazlov-sw-1#wr mem
Building configuration...
[OK]
```

Рис. 5: Настройка коммутатора 42-го квартала

Рабочая станция 42-го квартала

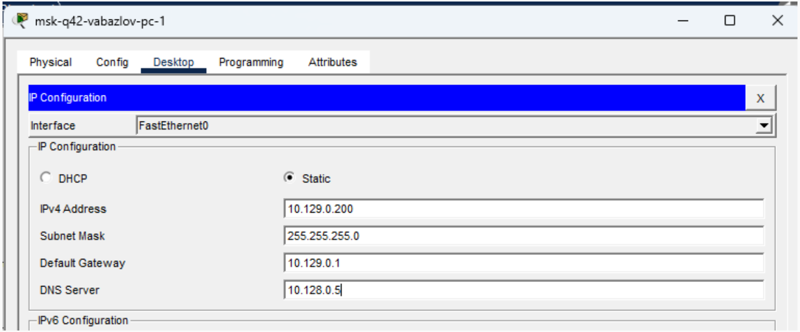


Рис. 6: IP-настройка ПК 42-го квартала

Настройка сети общежития

Шлюз общежития

```
msk-hostel-vabazlov-gw-1(config)#
msk-hostel-vabazlov-gw-1(config)#int g0/1
msk-hostel-vabazlov-gw-1(config-if)#sw trunk enc dot1q
msk-hostel-vabazlov-gw-1(config-if)#sw mode trunk

msk-hostel-vabazlov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-hostel-vabazlov-gw-1(config-if)#int f0/1
msk-hostel-vabazlov-gw-1(config-if)#sw trunk enc dot1q
msk-hostel-vabazlov-gw-1(config-if)#sw mode trunk

msk-hostel-vabazlov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-hostel-vabazlov-gw-1(config-if)#vlan 202
msk-hostel-vabazlov-gw-1(config-vlan)#name q42-management
msk-hostel-vabazlov-gw-1(config-vlan)#int vlan 202
msk-hostel-vabazlov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up

msk-hostel-vabazlov-gw-1(config-if)#no sh
msk-hostel-vabazlov-gw-1(config-if)#ip addr 10.129.1.2 255.255.255.0
msk-hostel-vabazlov-gw-1(config-if)#vlan 301
msk-hostel-vabazlov-gw-1(config-vlan)#name hostel-main
msk-hostel-vabazlov-gw-1(config-vlan)#int vlan 301
msk-hostel-vabazlov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-vabazlov-gw-1(config-if)#no sh
msk-hostel-vabazlov-gw-1(config-if)#ip addr 10.129.128.1 255.255.255.0
msk-hostel-vabazlov-gw-1(config-if)#exit
msk-hostel-vabazlov-gw-1(config)#ip routing
msk-hostel-vabazlov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.1
```

Коммутатор общежития

```
msk-hostel-vabazlov-sw-1>en
Password:
msk-hostel-vabazlov-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-vabazlov-sw-1(config)#
msk-hostel-vabazlov-sw-1(config)#
msk-hostel-vabazlov-sw-1(config)#
msk-hostel-vabazlov-sw-1(config)#int g0/1
msk-hostel-vabazlov-sw-1(config-if)#sw mode trunk
msk-hostel-vabazlov-sw-1(config-if)#int f0/1
msk-hostel-vabazlov-sw-1(config-if)#sw mode acc
msk-hostel-vabazlov-sw-1(config-if)#sw acc vlan 301
% Access VLAN does not exist. Creating vlan 301
msk-hostel-vabazlov-sw-1(config-if)#vlan 301
msk-hostel-vabazlov-sw-1(config-vlan)#name hostel-main
msk-hostel-vabazlov-sw-1(config-vlan)#int vlan 301
msk-hostel-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-vabazlov-sw-1(config-if)#no sh
msk-hostel-vabazlov-sw-1(config-if)#exit
msk-hostel-vabazlov-sw-1(config)#exit
msk-hostel-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-vabazlov-sw-1#wr mem
Building configuration...
[OK]
msk-hostel-vabazlov-sw-1#
```

Рис. 8: Настройка коммутатора общежития

Рабочая станция общежития

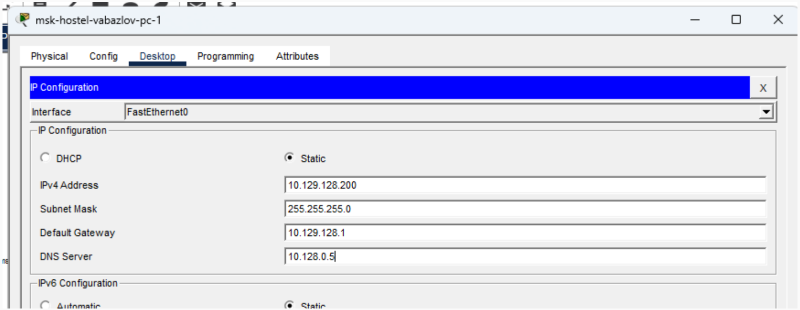


Рис. 9: IP-настройка ПК общежития

Настройка филиала в Сочи

Маршрутизатор филиала

```
sch-sochi-vabazlov-gw-1(config)#
sch-sochi-vabazlov-gw-1(config)#int f0/0
sch-sochi-vabazlov-gw-1(config-if)#no sh

sch-sochi-vabazlov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

sch-sochi-vabazlov-gw-1(config-if)#int f0/0.6
sch-sochi-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up

sch-sochi-vabazlov-gw-1(config-subif)#enc dot1q 6
sch-sochi-vabazlov-gw-1(config-subif)#ip addr 10.128.255.6 255.255.255.252
sch-sochi-vabazlov-gw-1(config-subif)#descr donskaya
sch-sochi-vabazlov-gw-1(config-subif)#
sch-sochi-vabazlov-gw-1(config-subif)#int f0/0.401
sch-sochi-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

sch-sochi-vabazlov-gw-1(config-subif)#enc dot1q 401
sch-sochi-vabazlov-gw-1(config-subif)#ip addr 10.130.0.1 255.255.255.0
sch-sochi-vabazlov-gw-1(config-subif)#desc sochi-main
sch-sochi-vabazlov-gw-1(config-subif)#int f0/0.402
sch-sochi-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.402, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up

sch-sochi-vabazlov-gw-1(config-subif)#enc dot1q 402
sch-sochi-vabazlov-gw-1(config-subif)#ip addr 10.130.1.1 255.255.255.0
sch-sochi-vabazlov-gw-1(config-subif)#desc sochi-management
sch-sochi-vabazlov-gw-1(config-subif)#exit
sch-sochi-vabazlov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
sch-sochi-vabazlov-gw-1(config)#exit
sch-sochi-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-vabazlov-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-vabazlov-gw-1#
```

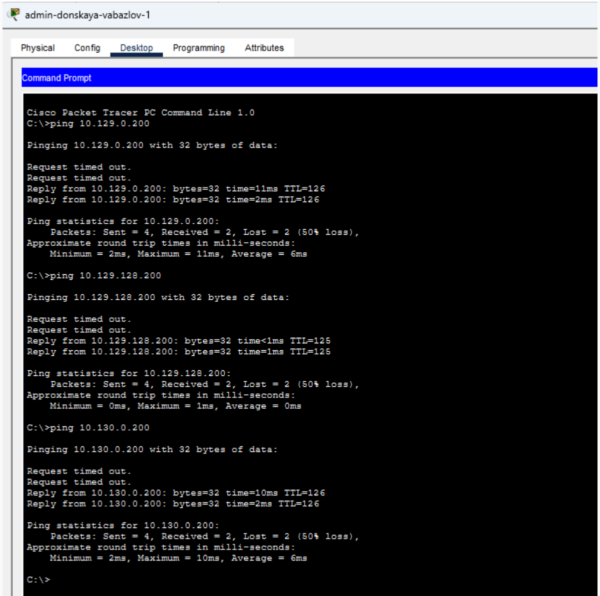
Коммутатор филиала

```
Password:
sch-sochi-vabazlov-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-vabazlov-sw-1(config)#
sch-sochi-vabazlov-sw-1(config)#int f0/23
sch-sochi-vabazlov-sw-1(config-if)#sw mode trunk

sch-sochi-vabazlov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

sch-sochi-vabazlov-sw-1(config-if)#int f0/24
sch-sochi-vabazlov-sw-1(config-if)#sw mode trunk
sch-sochi-vabazlov-sw-1(config-if)#vlan 6
sch-sochi-vabazlov-sw-1(config-vlan)#name sochi
sch-sochi-vabazlov-sw-1(config-vlan)#int vlan 6
sch-sochi-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

sch-sochi-vabazlov-sw-1(config-if)#no sh
sch-sochi-vabazlov-sw-1(config-if)#int f0/1
sch-sochi-vabazlov-sw-1(config-if)#sw mode acc
sch-sochi-vabazlov-sw-1(config-if)#sw acc vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-sochi-vabazlov-sw-1(config-if)#vlan 401
sch-sochi-vabazlov-sw-1(config-vlan)#name sochi-main
sch-sochi-vabazlov-sw-1(config-vlan)#int vlan 401
sch-sochi-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

sch-sochi-vabazlov-sw-1(config-if)#no sh
sch-sochi-vabazlov-sw-1(config-if)#exit
sch-sochi-vabazlov-sw-1(config)#exit
sch-sochi-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-vabazlov-sw-1#wr mem
Building configuration...
[OK]
sch-sochi-vabazlov-sw-1#
```

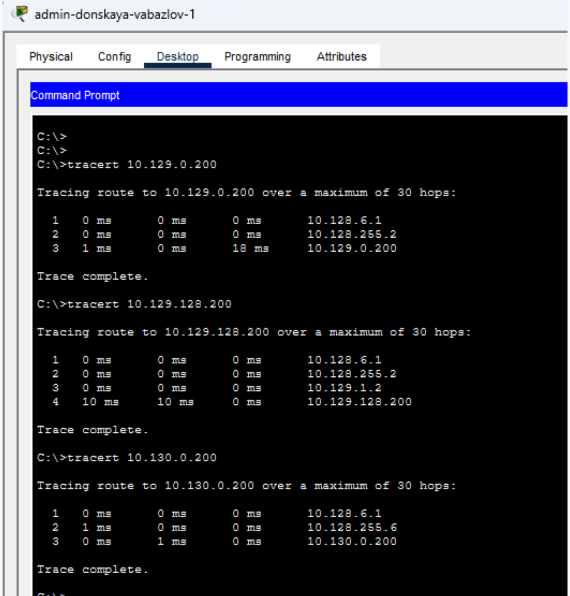
Рабочая станция филиала

Проверка связности

Проверка с помощью ping



Проверка маршрутов из центральной сети



Таблицы маршрутизации

Центральный шлюз

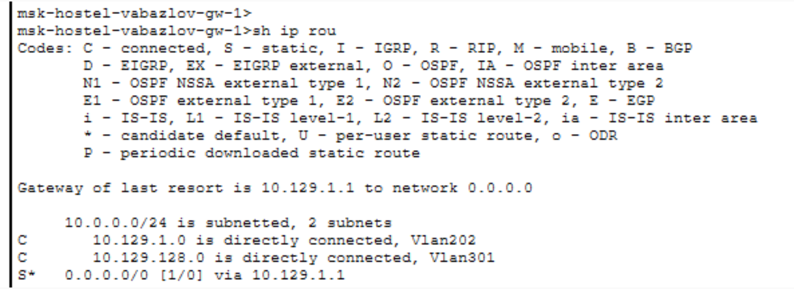
```
nsk-donskaya-vabazlov-gw-1>sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 18 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
L       10.128.6.1/32 is directly connected, FastEthernet0/0.104
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.1/32 is directly connected, FastEthernet0/1.5
C       10.128.255.4/30 is directly connected, FastEthernet0/1.6
L       10.128.255.5/32 is directly connected, FastEthernet0/1.6
S       10.129.0.0/16 [1/0] via 10.128.255.2
S       10.130.0.0/16 [1/0] via 10.128.255.6
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/28 is directly connected, FastEthernet0/1.4
L       198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*   0.0.0.0/0 [1/0] via 198.51.100.1

nsk-donskaya-vabazlov-gw-1>
```

Маршрутизатор 42-го квартала

```
msk-q42-vabazlov-gw-1>sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 7 subnets, 4 masks
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.2/32 is directly connected, FastEthernet0/1.5
C       10.129.0.0/24 is directly connected, FastEthernet0/0.201
L       10.129.0.1/32 is directly connected, FastEthernet0/0.201
C       10.129.1.0/24 is directly connected, FastEthernet1/0.202
L       10.129.1.1/32 is directly connected, FastEthernet1/0.202
S       10.129.128.0/17 [1/0] via 10.129.1.2
S*   0.0.0.0/0 [1/0] via 10.128.255.1
```

Рис. 15: Таблица маршрутизации 42-го квартала

Шлюз общежития



```
msk-hostel-vabazlov-gw-1>
msk-hostel-vabazlov-gw-1>sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

     10.0.0.0/24 is subnetted, 2 subnets
C       10.129.1.0 is directly connected, Vlan202
C       10.129.128.0 is directly connected, Vlan301
S*   0.0.0.0/0 [1/0] via 10.129.1.1
```

Рис. 16: Таблица маршрутизации общежития

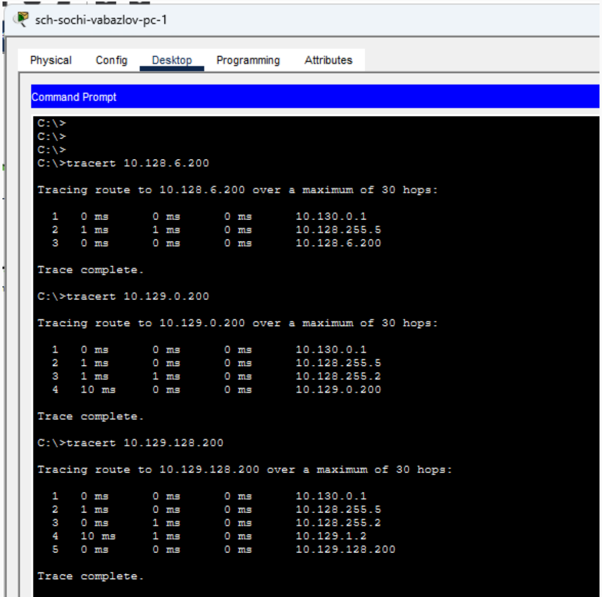
Маршрутизатор филиала в Сочи

```
sch-sochi-vabazlov-gw-1#
sch-sochi-vabazlov-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C       10.128.255.4/30 is directly connected, FastEthernet0/0.6
L       10.128.255.6/32 is directly connected, FastEthernet0/0.6
C       10.130.0.0/24 is directly connected, FastEthernet0/0.401
L       10.130.0.1/32 is directly connected, FastEthernet0/0.401
C       10.130.1.0/24 is directly connected, FastEthernet0/0.402
L       10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*   0.0.0.0/0 [1/0] via 10.128.255.5
```

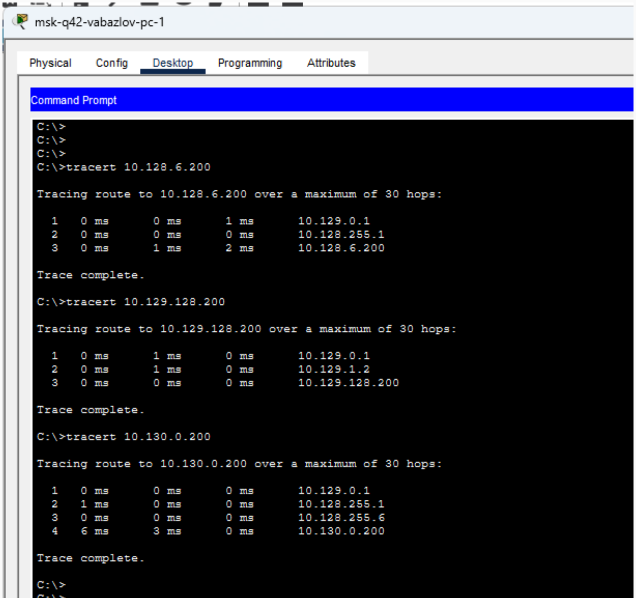
Рис. 17: Таблица маршрутизации филиала

Проверка трассировки

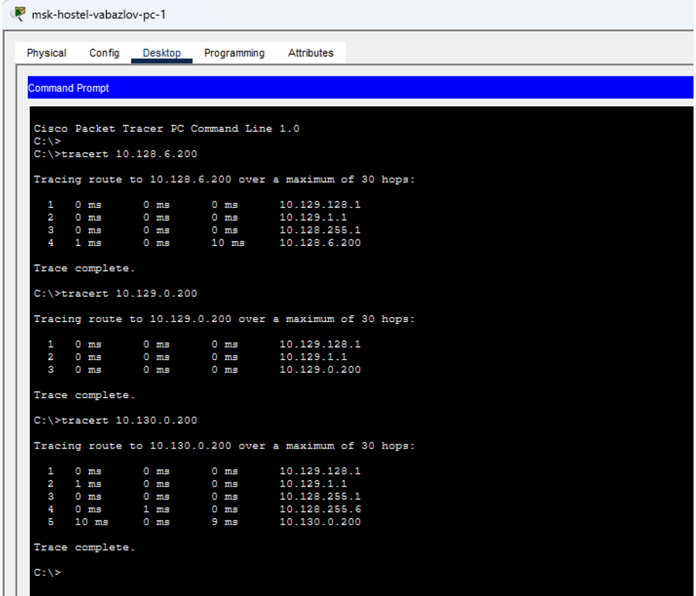
Трассировка из филиала в Сочи



Трассировка из сети 42-го квартала



Трассировка из сети общежития



Итоги работы

Вывод

В ходе лабораторной работы:

- собрана топология сети организации;
- настроены VLAN и trunk-соединения;
- созданы подинтерфейсы с инкапсуляцией 802.1Q;
- настроена статическая маршрутизация;
- заданы IP-параметры рабочих станций;
- проверена доступность узлов с помощью `ping`;
- проверено прохождение маршрутов с помощью `tracert`;
- просмотрены таблицы маршрутизации командой `show ip route`.

Настройка выполнена успешно: центральная сеть, сеть 42-го квартала, сеть общежития и филиал в Сочи взаимодействуют через сеть провайдера.